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MILITARY SPECIFICATION

PU<PS, POSITIVE DISPLACEMENT, MAIN/PORT USE,

MAIN BOILER FUEL SERVICE,

NAVAL SHIPBOARD) USE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

l.l Scope. This specification covers rotary positive displacement fuel service pumps, including prime movers, for main boiler service on Naval ships.

1.2 Classification. Pumps shall be of the following styles and classes, as specified (see 6.2.1):

Style I - Horizontal pumps.
Style II - Vertical pumps.
Class R-1 - Pumps for system gage pressures of 1000 pounds per square inch (lb/in^2) nominal.
Class R-2 - Pumps for system, gage pressures of 600 lb/in^2 nominal.
Class R-3 - Pumps for system gage pressures of 400 lb/in^2 norminal.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
FF-B-171 - Bearings, Ball, Annular (General Purpose).
FF-B-185 - Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187 - Bearings, Roller, Tapered.
FF-S-92 - Screw, Machine: Slotted, Cross-Recessed or Hexagon Head.
HH-P-96 - Paper, Gasket, Fiber (Animal or Plant) Sheet.
QQ-C-390 - Copper Alloy Casting (Including Cast Bar).
QQ-I-652 - Iron Castings, Gray.
QQ-T-390 - Tin Alloy Ingots and Castings, and Lead Alloy Ingots and Castings (Antifriction Metal) for Bearing Applications.
QQ-T-570 - Tool Steel, Alloy.
TT-P-645 - Primer, Paint, Zinc-Chromate, Alkyd Type.
GGG-P-701 - Puller, Mechanical Puller Attachment, Mechanical, and Puller Set, Mechanical.

MILITARY
MIL-S-869 - Steel Bars, Billets and Forgings-Alloy Nitriding Application.
MIL-S-890 - Steel: Forgings and Bars for Hulls, Engines, and Ordnance (Heat Treated).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

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MILITARY (continued)

MIL-S-901 - Shock Tests, H.I. (High Impact) ; Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-S-1222 - Studs, Bolts, Hex Tap Screws, and Nuts.
MIL-C-2212 - controllers, Electric Motor, A.C. or D.C. and Associated Switching Devices, Naval Shipboard.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-P-15024 - Plates , Tags and Bands for Identification of Equipment,
MIL-P-15024/5 - Plates, Identification.
MIL-S-15083 - Steel, Castings.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-P-15328 - Primer (Wash) , Pretreatment, Blue (Formula No. 117-B for Metals).
MIL-P-16789 - Packaging of Pumps, Including Prime Movers and Associated Repair Parts.
MIL-F-16884 - Fuel Oil, Diesel, Marine.
MIL-M-17060 - Motors, 60 Hertz, Alternating-Current, Integral HP (Shipboard Use) .
MIL-L-17331 - Lubricating Oil, Steam Turbine (noncorrosive).
MIL-M-17413 - Motor, Direct-Current, Integral HP (Shipboard Use).
MIL-M-17508 - Mounts, Resilient, Types 6E2000), 6E900, 6E900BB, 7E450, 7E450BB, 6E150, and 6E100.
MIL-T-17523 - Turbine, Steam, Auxiliary (and Reduction Gear) Mechanical Drive.
MIL-G-17859 - Gear Assembly, Propulsion (Naval) Shipboard Use) ,
MIL-G-18916 - Governors, Steam Driven Pump, Pressure Regulating, Naval Shipboard.
MIL-T-20157 - Tube and Pipe, Carborn Steel, Seamless.
MIL-S-22473 - Sealing, Locking, and Retaining Compounds , Single-Component .
MIL-I-24137 - Iron Castings, Nodular Graphitic (Ductile Iron) and Modular Graphitic (Corrosion Resisting Austenitic, Low Magnetic Permeability) (for Shipboard Application) .
MIL-M-24365 - Maintenance Engineering Analysis- Establishment of, and Procedures and Formats for Associated Documentation, General Specification For.
MIL-P-24377 - Packing Material, Asbestos, Braided, Impregnated With TFF (Polytretrafluoroethylene) , Surface Lubricated.
DOD-G-24508 - Grease, High Performance, Multi-Purpose (Metric),
MIL-I-45208 - Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-167-1 - Mechanical Vibration of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278 - Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment
MIL-STD-777 - Schedule of Piping. Valves. Fittings, and Associated Piping Components for Surface Ships.
MIL-STD-1552 - Provisioning Technical Documentation, Uniform DOD Requirements for.
MIL-STD-1561 - Provisioning Procedures, Uniform DOD.

(Copies of specifications standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer) .

2.2 Other publications . The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents Government Printing Office, Washington, DC 20402.)

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HYDRAULIC INSTITUTE
Test Code of Standards - Rotary Pump Section.

(Application for copies should be addressed to the Hydraulic Institute, 2130 Keith Building, Cleveland, OH 44115.)

AMERICAN IRON AND STEEL INSTITUTE (AISI)
Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 1000 16th Street, NW, Washington, DC 20036.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
B18.2.1 - Square and Hex Bolts and Screws, Including Askew Head Bolts, Hex Cap Screws, and Lag Screws.
B18.2.2 - Square and Hex Nuts.
B18.6.3 - Slotted and Recessed Head Machine Screws and Machine Screw Nuts,
B46.1 - Surface Texture.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A27 - Mild- to Medium-Strength Carbon-Steel Castings for General Application.
A48 - Gray Iron Castings.
A193 - Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service.
A194 - Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High-Temperature Service.
A216 - Carbon-Steel Castings Suitable for Fusion Welding for High-Temperature Service.
A325 - High-Strength Bolts for Structural Steel Joints, Including Suitable Nuts and Plain Hardened Washers.
A395 - Ferritic Ductile Iron Pressure-Retaining Castings for Use at Elevated Temperatures .
A695 - Steel Bars, Carbon, Hoc-Rolled, Special Quality, for Fluid Power Applications .
B23 - White Metal Bearing Alloys (known commercially as “Babbitt Metal”) .
B584 - Copper Alloy Sand Castings for General Applications.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PP 19103,)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies) ,

3. REQUIREMENTS

3.1 Qualification, Pumps furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at time set for opening of bids (see 4.3 and 6.3).

3.2 Pump characteristics. General range of pump characteristics shall be in accordance with table I. Characteristics of pumps to be procured shall be as specified (see 6.2.1).

TABLE I. Pump characteristics.

Discharge gage pressure	Rated capacity	Suction lift at rated capacity 1/ 2/	Viscosity range of liquid for test purposes	Range of liquid temperatures under operating conditions 3/	Type of drive
(Lbs/in^2)	(Gal/min)	(Inches of mercury)	(SSU)	(°F)	
350-1150	5-100	POS-20	35-100	60-120	Geared turbine or motor

See footnotes at top of next page.

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1/ Referred to centerline of pump inlet.
2/ Suction lift: 5 inches of mercury (in Hg) - grade JP-5 fuel in accordance with MIL-T-5624.
10 in Hg - marine diesel fuel (DFM) in accordance with MIL-F-16884,
3/ Temperatures do not necessarily correspond with the viscosity range, but indicate temperature pumps may be expected to encounter in service.

3.3 General design. Design and construction of pumps shall comply with the space and weight requirements as specified (see 6.2.1). Pump unit shall perform under the list, pitch, roll, and trim conditions as specified (see 6.2.1).

3.3.1 Pumps shall be self-priming and capable of removing air from the suction lines, but shall be primed before initial startup.

3.3.1.1 Where the lubrication of the pumps is supplied by the fluid pumped, all parts of the pumps shall operate in accordance with the requirements of this specification when pumping these fluids.

3.3.1.2 Tolerances shown on a drawing shall be complied with during manufacture unless otherwise indicated by note, legend, or key.

3.3.2 Reliability. Users of and contractors to this specification are urged to communicate to the Naval Ship Engineering Center (NAVSEC) any find.ings related to the requirements or lack of requirements of this specification whereby improvement of equipment reliability can be achieved. Equipment specified herein shall be operated, maintained, and repaired onboard Navy ships. Attention is directed to the high equipment density of shipboard machinery spaces, to the fact that required maintenance and repair shall be capable of being made underway in heavy seas, that equipment shall operate unattended, and that maintenance personnel may not be experienced mechanics. The requirement for maximum reliability directly relates to those shipboard environmental and service conditions and they shall be fully considered in the pump design. Aspects of “human engineering” design shall also be fully explored and implemented in the equipment design to minimize the possibility of failure through improper operation and maintenance and to preclude personnel safety hazards. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable functionally and from considerations of reliability.

3.3.2.1 Positioning and alinement of parts in assembly shall employ positive means such as shoulder, tongue and groove, or other locating techniques whereby correct reassembly is repeatedly assured. In consonance with the concept of ease of maintenance and where not already specified, the pump designer shall consider the following features for incorporation in the design and technical documentation wherever such incorporation will significantly contribute to ease of maintenance without compromising performance reliability:

(a) Casing assembly guide pins.
(b) Alinement and solid positioning dowels.
(c) Jacking screws.
(d) Ease of access to couplings.
(e) Remakeable piping connections.
(f) Piping out of way of maintenance access.
(g) Ease of mechanical seal and bearing replacement.
(h) Guards for rotating and moving components.
(i) Warning plates to prevent casualties to equipment and personnel.
(j) Provisions for connecting instruments for performance evaluation.
(k) Attached instruments for monitoring performance.
(1) Visibility and access to attached instruments.
(m) Thoroughness of operating instructions.
(n) Thoroughness of preventive maintenance instructions.
(o) Explicit assembly and disassembly instructions.

3.3.2.2 Each pump shall be designed to have a service life of 30 years, of which 40 percent shall consist of actual operation. There shall be no limit on the number of starts during the life of the pump. It shall be assumed that during the life of the pump, parts subject to unavoidable wear and deterioration (with the exception of seals) shall require replacement at intervals not less than 3 years. Parts subject to wear, deterioration, and normally requiring replacement at 3 year intervals during the service life of the pump (with the exception of seals) shall be designed for a life of 10,000 actual operating hours pumping the fluid specified (see 6.2.1) and they shall be identified in the appropriate drawings and manuals. Requirements set forth above shall not be construed or interpreted as a warranty requirement nor otherwise affect the manufacturer’s warranty.

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3.3.3 Shock and vibration.

3.3.3.1 Pumps with drivers, appurtenances, and controls, shall be capable of passing a shock test in accordance with MIL-S-901, as specified in 4.4.3.8.

3.3.3.2 Shock mounts (resilient mounts for shock attenuation) shall not be used. When noise attenuation mounts are specified, pumps shall be designed to pass the shock test on standard Navy noise attenuation mounts in accordance with MTL-M-17508,

3.3.3.3 Bolts stressed in shear shall be installed in holes with a minimum of clearance. Hole diameters shall be no more than 1/32 inch larger than the bolts for sizes up to and including 3/4 inch, and no more than 1/16 inch larger than bolts of greater than 3/4 inch size. Mounting bolts for fastening the equipment shall conform to grade 2 or better of MIL-S-1222.

3.3.3,4 Pump units shall not be damaged or caused to malfunction either by internally excited vibrations, or by the environmental vibrations specified in MIL-STD-167-1. Frequencies shall be tested up to and including 33 hertz (Hz) .

3.3.3.5 When required (see 6.2.1), pumps shall be designed and tested to meet the airborne noise requirements of MIL-STD-740 and as specified herein. The contractor shall obtain noise measurements in accordance with the requirements, measurements, and data reporting procedures as contained in MIL-STD-740. Equipment shall not exceed the airborne noise criteria identified in table II.

TABLE II. Octave band sound pressure levels.

Octave band center frequency, in Hz									
	31.5	63	125	250	500	1K	2K	4K	8K
SPL	95	90	85	82	79	76	75	75	75

Note: Reference pressure is 0.0002 dynes/cm^2.

3.3.4 Mounting.

3.3.4.1 Pump units shall be horizontally or vertically mounted, as specified (see 6.2.1).

3.3.4.2 Each horizontal pump and driving unit shall be mounted on a common bedplate.

3.3.4.3 Each vertical unit shall be base mounted or table mounted, as specified (see 6.2.1).

3.3.4.4 A bracket shall be provided between the pump and driver of vertical units.

3.3.4.5 Bedplates, bases, table mounts, and brackets shall be sufficiently rigid to permit handling, shipment, and installation of the units onboard ship to minimize misalinement of the assembled units and such that the normal distortion, weaving, or vibration of the supporting structures onboard ship shall not cause misalinement betweeen the pumps and driving units. The driver shall be capable of being doweled and the unit realined by the installing activity after final installation. The contractor shall furnish a warning plate advising of the above requirement.

3.3.4.6 Each component part of an assembled flexible coupled unit supported directly by a bedplate shall be doweled thereto to facilitate reassembly and maintenance of alinement except that final doweling of the driver shall be performed by the installing activity after final installation.

3.3.4.7 Bearing and seating surfaces of bedplates, bases, table mounts, and brackets shall be machine finished.

3.3.4.8 Equipment shall not be rigidly supported from more than one plane.

3.3.4.9 In the event special bulkhead mounting is required, sideplates shall be furnished subject to the conditions specified for baseplates (see 3.3.4.1 through 3.3.4.8) .

3.3.4.10 Each vertical unit, in order to facilitate interchangeability, shall be constructed to permit rotation of the pump with respect to its driver. Eight or more positions of pump flanges relative to the driver shall be available.

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3.3.4.11 Bedplates, bases, table mounts, and brackets shall be cast steel or structural steel fabricated by welding.

3.3.4.12 Vertical pumps shall be base mounted or center of gravity mounted. The base and support bracket shall have provisions for bolting to foundation.

3.3.4.13 Eyebolts, lugs, holes, or other means shall be provided to permit attachment of lifting gear for lifting the assembled pump, driver, and attached accessories as a complete unit.

3.3.5 Pump bearings.

3.3.5.1 Radial and thrust bearings shall be of the hydrodynamic or rolling contact type.

3.3.5.2 Rolling contact bearings shall be of the ball type, in accordance with FF-B-171. Roller bearings may be recommended if use of ball bearings is Impractical. Rolling contact bearings shall be selected to result in a minimum B-10 life of 10,000 hours, calculated in accordance with FF-B-171, FF-B-105, or FF-B-187.

3.3.5.3 Hydrodynamic radial bearings may be of the fixed or self alining type, and shall operate under full fluid film lubrication for all operating conditions, except boundary lubrication may be used during pump start-up and shut-down. Where the requirement for full fluid film operation cannot be met, a hydrostatic bearing may be recommended by the manufacturer. The hydrostatic bearing shall operate without metal-to-metal contact over the entire range of the pump characteristic curve.

3.3.5.4 Hydrodynamic and hydrostatic bearing surface and the journal shall be finished to 32 roughness height rating (RHR) or better.

3.3.5.5 Bearing installation.

3.3.5.5.1 Rolling contact bearings, where used, shall be installed external to pumping area between coupling and mechanical seal.

3.3.5.5.2 Sleeve bearings shall be split along the axis and shall be arranged to permit renewal or refitting without removal of the pump rotor from the casing. They shall be fitted in their housing seat under a light press fit and shall have locking-lips or dowels to prevent circumferential rotation and axial movement. Shells and their seatings shall be machine finished.

3.3.5.5.3 Bearing housings shall be cast integral with the bracket or secured thereto in such a manner as to insure alinement, Bearing housing to bracket seats shall be machined such that the precision of bearing alinement is insured.

3.3.5.5.4 Bearing brackets shall be either cast. integral with the casing, securely bolted to the casing on a machined shoulder, or held in alinement with the casing by at least two heavy dowels and securely bolted thereto. The use of bolts alone for securing brackets is prohibited.

3.3.5.6 Thrust bearings.

3.3.5.6,1 A thrust bearing shall be installed on each pump for counteracting any unbalanced hydraulic or mechanical thrust in either direction. In this connection consideration shall be given to the fact that rolling, pitching, or list of a ship at sea may introduce thrust loads even though the unit is in hydraulic balance.

3.3.5.6.2 Thrust bearings shall be of the straight ball or roller type, combined radial and thrust ball or roller type, or plain sliding surface type.

3.3.5.6.3 Thrust bearings of the ball or roller type and their installation shall be in accordance with FF-B-171 or FF-B-185, as applicable.

3.3.5.7 Bearing lubrication.

3.3.5.7.1 Lubrication of rolling contact bearings shall be in accordance with FF-B-171 or FF-B-185.

3.3.5.7.2 Rolling contact bearing lubrication method shall prevent bearing failure resulting from overgreasing.

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3.3.5.7.3 Rolling contact bearings having a value in excess of 200,000 shall be oil lubricated unless the manufacturer can demonstrate to the design review agency that the specific grease lubricated bearing application has been proven in previous Navy shipboard service. Grease lubrication is preferred when nd_m is less than 200,000.

Where:

n = Revolutions per minute (r/min).

d_m = Mean diameter $\frac{OD - ID}{2}$.

3.3.5.7.4 Where forced feed lubrication is provided for turbine-driven pumps, the lubricating system shall be in accordance with requirements of the applicable turbine specification (see 3.5.1). Lubricants shall be in accordance with the following: grease shall conform to DOD-G-24508 and oil shall conform to MIL-L-17331. Bearings shall be capable of being relubricated without removing the bearings and shall not require frequent lubrication if the pump is not operated over an extended period of time.

3.3.5.7.5 Thrust bearing housings shall be designed so that the thrust bearings will be adequately lubricated immediately when the shaft starts to turn. This shall be accomplished by properly locating oil supply and drain pipes, and by adequate sealing of the bearing housings so that an adequate supply of oil will remain in the bearing housings,

3.3.6 Piping and valves.

3.3,6.1 Valves, flanges, fittings, and bolting shall conform to MIL-STD-777 for surface ships and as specified (see 6,2.1) , except tapered pipe threads are acceptable for use in grease fittings located on nonmoving parts.

3.3.6.2 Gland seal piping/tubing shall be in accordance with category E-1 of MIL-STD-777. Related valves/fittings shall be steel in accordance with MIL-T-20157. Gland seal piping/tubing shall be sized as required by pump flow rate and all connections and fittings shall be screw type.

3.3.7 Casings and heads.

3.3.7.1 Casings, also referred to as cylinders, are those parts which enclose the rotors. Bores of the casings shall be parallel to the rotors.

3.3.7.2 Casings shall be fitted with casinq liners, also referred to as rotor housings, and shall be replaceable wearing parts located between casing bore and rotors.

3.3,7.3 Suction and discharge openings shall be cast integral with the casings or heads, shall have machined and scraped or ground flange faces in accordance with ANSI B46.1, and shall be located as specified by NAVSEC or the procuring agency concerned (see 6.2.1).

3.3.7.4 Heads shall have accurately machined faces in accordance with ANSI B46.1, Joints between casings and heads may be made up either metal-to-metal or with a thin plant or animal fiber gasket in accordance with HP-P-96 not exceeding 0.010 inch in thickness. Heads shall be rabbetted or doweled to the casings.

3.3.7.5 Forcing bolts shall be provided in pumps for breaking joints.

3.3.7,6 Clearance shall be provided around bolt heads and nuts to permit the use of ordinary tools.

3.3.7.7 Fitted bolts or heavy dowel pins shall be provided in flexible-coupled pumps to insure maintenance of alinement in reassembly. Tapered dowel pins shall be corrosion-resistant, shall have a threaded end and nut design, and shall be secured against coming adrift under shock loading.

3.3.8 Rotors and shafts.

3.3.8.1 Rotors shall be accurately machine finished or ground throughout. They shall be integral with or secured to the shafts by keys. Rotors shall be secured positively against lateral displacement on shafts.

3.3.8.2 Shafts shall have a minimum hardness of 275 Brinell.

3.3.8.3 Except where locknuts are used, shaft threading shall be opposite to direction of rotation.

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3 .3.9 Timing gears.

3.3.9.1 Timing gears where used shall be of the spur or double helical type. Gears shall be fully enclosed and adequately lubricated. Timing gears shall be secured to the shafts so as to prevent either radial or lateral movement.

3.3,9.2 Provision shall be made for replacement of timing gears in the field, so that it will be satisfactory to carry and replace individual parts or rotating elements. The method of installing and timing the timing gears shall be thoroughly explained, with sketches, if necessary, in the technical manuals.

3.3.10 Mechanical shaft seals.

3.3.10.1 Pumps shall be equipped with mechanical shaft seals. Magnetic type seals shall not be used.

3.3.10.2 In addition to the requirement for mechanical shaft seals, pump shall have a backup stuffing box to accommodate two or more rings of conventional or “soft” packing for emergency use in the event of mechanical shaft seal failure. Packing shall be teflon impregnated asbestos in accordance with MIL-P-24377.

3.3.10.3 Packing rings shall be inserted without removing the mechanical shaft seal.

3.3.10.4 Mechanical shaft seals shall be positioned on the shaft by means of stub or step sleeves. Mechanical shaft seals shall not be positioned by use of set screws.

3.3.10.5 Positive liquid pressure shall be supplied to the seal faces under all conditions of operation and there shall be adequate circulation of the liquid at the seal faces to minimize the deposit of foreign matter on the seal parts. Pump certification data shall contain certification by the seal manufacturer that there will be adequate circulation of liquid and adequate lubrication at the seal faces when the seal is installed as shown on the drawing.

3.3.10.6 The design shall be such as to insure that if seal leakage occurs, fluid cannot reach bearings or drive units. A fluid leakage collection reservoir with means for adequate drainage to a safe point shall be provided to control any leakage that might occur.

3.3.11 Couplings

3.3.11.1 A flexible coupling of all-metal construction shall be installed between the pump and the driving unit.

3.3.11.2 Flexible coupling pump hubs shall be keyed to the shafts and secured by lock nuts . For pump shafts l-inch in diameter and larger the hubs shall he fitted on a taper with keys.

3.3.11.3 Horizontal flexibly coupled units with bedplates shall have coupling guards provided. Guards shall permit ready access to the coupling for lubrication and inspection.

3.3.12 Rubbing speeds.

3.3.12.1 Maximum rubbing speed in feet per minute (ft/min) for a particular class pump shall be the speed established for the class pump during the qualification test. Pumps shall not be offered or furnished on any contract or order which operate at a rubbing speed greater than the rubbing speed at which the pump of the same class operated on qualification test.

3.3.13 Threaded fasteners.

3.3.13.1 Threaded parts, such as bolts, studs, and nuts shall conform to Handbook H28. The usc of tap bolts or cap screws is prohibited. Setting end of studs shall be class 5 fit, or class 3 fit with locking resin, in accordance with MIL-S-22473. Nut end shall be class 3 fit.

3.3.14 Welding Welding shall be in accordance with MIL-STD-270.

3.3.15 Painting.

3.3.15.1 External unmachined surfaces of ungalvanized ferrous metal parts shall be thoroughly cleaned and coated with one coat of pretreatment conforming to MIL-P-15328, and one coat of zinc chromate primer conforming to TT-P-645, followed by a finish coat of light gray equipment enamel conforming to type I, class 2 of MIL-E-15090.

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3.3.15.2 Painting of external surfaces of nonferrous parts of pumps will not be required.

3.3.16 Identification plates.

3.3.16.1 Identification plates shall be furnished on each pump. Plates shall be type A or B in accordance with MIL-P-15024 and MIL-P-15024/5, except that identification plates of plastic or aluminum are not acceptable.

3.3.16.2 Identification plates shall be secured to equipment with corrosion-resistant metallic screws or rivets.

3.3.16.3 Pump identification plates shall contain data as follows:

(a) Manufacturer's name.
(b) Manufacturer's model style, class, and size.
(c) Service application.
(d) Manufacturer's serial number.
(e) Salient design characteristics:
  (1) Capacity in gallons per minute (gal/rein).
  (2) Discharge pressure.
  (3) Suction lift.
  (4) Speed of shaft in r/rein.
  (5) Brake horsepower (hp).
  (6) Test pressure.
  (7) Viscosity range (operating).
(f) Contract number (and item number of multiple unit orders).
(g) National stock number (NSN).
(h) Section for Defense Contract Administration Service (DCAS) stamp.
(i) NAVSHIPS technical manual number.

3.3.16.4 Each driving unit and each accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.4 Materials.

3.4.1 Materials of the pump shall conform to the materials specified in table III. However, this specification is not intended to be restrictive provided proposed alternate materials will give equal or better service than the materials specified. Components of the pump for which specific materials are not specified shall be materials compatible for the service intended. Materials which can be sensitized and are subjected to heat treatment in the sensitization range during fabrication shall be capable of passing the intergranular corrosion tests of the base material specifications. Particular attention should be given to avoid sensitization of materials during hard facing, stress relieving, or repair welding. Use of cadmium-plated parts and fasteners, including washers, is prohibited. Items and parts used in the pump shall be new and unused. Reclaimed material shall be used to the maximum extent practicable in producing such items and parts provided that the pump and items/parts conform to the requirements of this specification.

TABLE III. Materials for pump parts.

Application[1/]	Material	Applicable document[2/]
Casing with liner and heads	Cast steel	MIL-S-15083, grade B or ASTM A216, grade WCA or ASTM A27, grade U60-30
	Gun metal	QQ-C-390, alloy 903 or ASTM B584, alloy 903
	Nodular iron	MIL-I-24137, class A or ASTM A395
Shafts	Plain carbon and low alloy steel	MIL-S-890 or ASTM A695, type C
	Steel (nitriding grade)	MIL-S-869, class A or B

See footnotes at end of table.

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TABLE III. Materials for pump parts. - Continued

Application[1]	Material	Applicable document[2]
Rotors	Plain carbon and low alloy steel	MIL-S-890 or ASTM A695, type C
	Steel (nitriding grade)	MIL-S-869, class A or B
	High leaded tin bronze	ASTM B584, alloy 937 or 943 or QQ-C-390, alloy 937 or 943
	Tool steel	QQ-T-570, class D-2 or AISI D-2
Sleeve bearings	Bearing bronze	QQ-C-390, alloy 932, 934, or 937 or ASTM B584, alloy 932 or 937
	Leaded tin bronze	QQ-C-390, alloy 923 or ASTM B584, alloy 923
	Carbon	High strength mechanical carbon graphite (bonded and copper-lead treated)
	Gray iron casting	QQ-I-652 or ASTM A48
Rotor housings or cylinder liners	Bearing bronze	QQ-C-390, alloys 932, 934, or 937 or ASTM B584, alloy 932 or 937
	Steel with babbitt	QQ-T-390, grade 2 or ASTM B23, alloy 2
Timing gears	Hardening grade steel	Manufacturers standard: Minimum surface hardness 52 Rc Minimum effective case depth 0.015 inch
Casing side wearing plates	Chromium-moly-cobalt iron base hard facing alloy	Haynes alloy no. 93, or equal
	Nitriding grade steel	MIL-S-869, class A or B
	Cast iron	QQ-I-652 or ASTM A48
Bolts, studs and nuts, hexagon head[3]	Steel	MIL-S-1222 or ANSI B18.2.1 and B18.2.2 or ASTM A193, grade B7 or ASTM A194, grade 2H or ASTM A325
Machine screws, slotted or cross recessed	Steel	FF-S-92 or ANSI B18.6.3

[1] Hardware or subassembly parts which are used in a major pump part listed in the table and which perform the same or similar functions as the major part (e.g., casing drain plug) shall be of the material specified for the major part.

[2] Detail drawings and lists of material shall reference the specification followed in each case (preferably in the order specified in 3.7.1.4.1), and shall include the class, type, or grade of material used in each case, as applicable.

[3] The design of bolts, studs, and nuts shall be in accordance with MIL-S-1222. Nickel-copper-aluminum alloy bar stock used for threaded fasteners shall be as specified in MIL-S-1222.

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3.5 Pump drivers. The type of pump driver shall be as specified (see 6.2.1).

3.5.1 Turbines. turbines for driving pumps shall be in accordance with MIL-T-17523, as specified (see 6.2.1).

3.5.2 Regulating governors.

3.5.2.1 Unless otherwise specified (see 6.2.1) , one constant pressure regulating governor shall be furnished with each turbine-driven pump for flushing service. Pressure regulating governor shall be in accordance with MIL-G-18916 and of the type and characteristics, as specified (see 6.2.1).

3.5.2.2 Steam pressure drop through pump pressure regulating governors should be considered when rating turbine driven pumps.

3.5.3 Reduction gears,

3.5.3.1 Where reduction gears are used, they shall be of the spur, bevel, single or double helical, or worm type in accordance with MIL-G-17859.

3,5.3.2 Reduction gears shall be independent of the pump and connected thereto by a flexible coupling. Reduction gears may, however, be an integral part of the driving unit.

3.5.4 Motors.

3.5.4.1 Motor drives shall be of the type and characteristics, as specified (see 6.2.1) and shall conform to MIL-M-17413 for direct current (d.c.) motors, and to MIL-M-17060 for alternating current (a,c.) motors. Unless otherwise specified (see 6.2.1), motors shall have the following features:

Service	A
Ambient temperature	50°C
Duty	Continuous
Enclosure	Totally enclosed or totally enclosed fan-cooled
Insulation	Class B, B, II, or N
Temperature rise	Not to exceed class B
Type of bearings	Ball, type III, class 2, in accordance with FF-B-171
Winding:	
D.C.	Shunt
A.C.	Squirrel-cage induction
Design, a.c.	Design B or C

The full load torque at all speeds of multi-speed motors shall be constant.

3.5.4.2 Motor rating, The hp rating of each motor shall be not less than that required by the pump under maximum power conditions as determine? by actual tests of identical or similar-eqGipment. If such data is not available, the required hp rating of the motor shall be determined by the following formula:

$$\text{Hp of driver} = \frac{1.05 \text{ (maximum hp output of the pump)}}{\text{(efficiency of pump)}}$$

Actual motor rating shall be in accordance with Navy standard motor ratings.

3.5.5 Controllers.

3.5.5.1 Motor controllers and pressure switches shall be in accordance with MIL-C-2212 for d.c. and a.c. Unless otherwise specified (see 6.2.1), controllers shall have the following features:

Operation	Magnetic
Type:	
D.c.	Resistor starting
A.C.	Across-the-l~ne starting
Performance	Semi-automatic
Function	Motor starting
Duty	Continuous

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Enclosure:	
Controller	Dripproof
Master switches	Spraytight
Protective features	Thermal overload and low voltage protection
Master switch	Remote START-BYPASS-RUN-STOP selector Remote speed selector switch (if multi-speed motors are used) Remote pressure switch.

3.5.5.2 Operation of motor controllers shall be semi-automatic except that a remote pressure switch shall be installed in the control circuit such that the motor will automatically stop on reduction of pump discharge pressure below a certain predetermined pressure. No automatic starting feature shall be provieded. The START-BYPASS-RUN-STOP selector shall have 3 positions. The STOP and RUN shall be maintained positions, while the START momentary contact shall be spring-rcturned to the RUN position. The START contacts shall bypass the pressure switch contacts until the pump builds up to close the pressure switch.

3.6 Onboard repair parts and tools.

3.6.1 Onboard repair parts. Repair parts (onboard) shall be furnished in accordance with MIL-STD-1552 and MIL-STD-1561.

3.6.2 Tools.

3.6.2.1 Special tools required for the maintenance and repair of the pump units shall be furnished. Special tools are defined as those not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the DCAS). The number of sets of special tools shall be as specified (see 6.2.1) .

3.6.2.2 Tools which are available in the National Supply Catalog, such as common wrenches and standard pullers, will not be required. Standard complete or limited sets of pullers or parts of puller sets in accordance with GGC-P-781 shall be identified as to type and use in notes on the outline or section assembly drawings. Special tools shall be detailed, and included in the list of materials.

3.-? Technical data. The contractor shall prepare drawings, certification data, and manuals, in accordance with the data ordering documents included in the contract or order (see 6.2.2) and as specified in 3.7.1 through 3.7.2.4).

3.7.1 Drawings In addition to the drawing content required by the data ordering document (see 6.2.2), the unique features specified in 3.7.1.1 through 3.7.1.6 shall be included.

3.7.1.1 General requirements.

3.7.1.1.1 "[manufacturer's use only" notes. Information intended for "manufacturer's use only" shall be so designated. Conversely, on a commercial drawing with wide usage, it is permissible to designate portions thereon, "For Navy use only".

3.7.1.2 Design drawings. Design drawings shall consist of the drawings described in 3.7.1.2.1 through 3.7.1.2.4.2.

3,7.1.2.1 Sectional assembly drawings. Sectional assembly drawings shall include a sectional assembly with references to the list of material for identification of individual parts . Side or partial views may be required in order to show parts not otherwise shown in the main section view. Running clearances shall be shown and shall be dimensioned as radial clearances . (The radial clearance equals one half the diametrical clearance.) Tightening torques and thread lubrication requirements for threaded fasteners shall be shown on the assembly drawing.

3.7.1.2.2 List of material. List of material shall include every part required in the pump assembly, including those parts not required to be detail.ed. List of material shall include an indication of each part required by this specification to be furnished as an onboard repair part (this is not a repair parts list) . List of material shall include the manufacturer's drawing number and service part number for parts for which detail drawings are required (see 3.7.1.2.4) as well as quantity required, material type, and material specification for parts. Parts shall be named to indicate the function they serve. Parts for which detail drawings are not required shall be identified as to dimensions.

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3.7.1.2.3 Detail drawings s. Detail drawings shall be furnished of parts and subassemblies necessary for evaluation of the equipment, and of parts necessary for maintenance and overhaul of the pumps. Details of these parts shall be complete in every respect to permit emergency manufacture by a Government activity without assistance from the original manufacture. Details of pump casings shall be furnished, but the dimensioned development of cored hydraulic passages need not be shown. Subassemblies whose parts cannot be procured or serviced individually, should be shown as a single part and so indicated. Mulidetail drawings are preferred, but monodetail drawings may be used.

3.7.1.2,4 Outline drawings and certification data (see 6.2.2).

3.7.1.2.4.1 Outline drawings and certification data shall be furnished as supplemental drawings to sectional assembly and detail drawings. Separate outline drawings, drawing list, and certification data shall be furnished under each contract or order unless the complete equipments covered by the outline drawing and the referenced drawings are in fact identical in all respects.

3.7.1.2.4.2 Outline drawings, in addition to the certification data, shall contain the following:

(a) Dimensional outline assembly drawing of the pump with its prime mover, bedplate, and attached auxiliaries.
(b) Complete performance data of pump, prime mover, and attached auxiliaries, if applicable.
(c) Complete equipment performance curves, based on actual tests (see 4.4.2.1) (the original submission may show design performance curves vice test curves) .
(d) Table of weights of individual components, and weight of complete unit.
(e) Shipbuilder's connections showing size, type, and dimensions of flanges.
(f) Center of gravity of the pump component and of the complete assembly,
(g) Radii of gyration of complete assembly about each of the three principal axes.
(h) Identification of system in which installed (if known) .
(i) Endurance test, high impact shock tests, and nondestructive tests (if performed) .
(j) In addition, the outline drawing shall include overall dimensions of the complete unit including locating dimensions of fittings and connections, the space required for removal and replacement of parts for maintenance) and mounting information.

3.7.1.3 Basic design drawings. Basic design drawings covering pumps under this specification are intended for submittal to NAVSEC for review prior to and independent of invitations for bids . As such, they are different from design drawings as described in 3.7.1.2. Each drawing may cover a range of sizes of pumps so long as they are of the same basic design.

3.7.1.3.1 Contents of basic design drawings. These drawings shall consist of small scale (not necessarily to scale plan and evaluation views showing overall dimensions. If the plans cover a family of pump sizes of the same design, the dimensions may be tabulated. The range of capacities, pressure, and speeds shall he shown. The main part of these drawings shall be an undimensioned sectional assembly with complete list of material which shall include every part required in the pump assembly. If materials of some parts may change with various pump characteristics or with differing fluids to be pumped, a table shall be included to show these variations. Alternate subassembly arrangement of different parts may be included as desired.

3.7.1.3.2 Detail drawings or individual parts are not required in connection with basic design drawings, but may be included if desired by the manufacturer for clarification, or to obtain advance comments of NAVSEC.

3.7.1.4 Material identification,

3.7.1.4.1 Preferred material reference. Where materials of identical or equal quality can be identified by more than one specification or standard, the drawings need reference only one such specification or standard. In selecting the specification or standard to be referenced, the following is the order of preference:

(a) Federal specification.
(b) Military specification.
(c) Industry and Technical Society specification or standard.
(d) Manufacturer's specification or standard.

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3.7.1.4.2 Material substitutions. Where materials other than covered by (a), (b), and (c) of 3.7.1.4.1 are referenced and by approved by design review agency, the drawings shall show the complete chemical and physical properties of the approved material. In addition, the drawings shall identify the material in terms of the nearest Federal or Military specification, in order to enable Naval repair facilities to make emergency repair as necessary from Navy material stocks.

3.7.1.5 Drawing identification. Prime contractors who purchase items from subcontractors shall use the subcontractor's drawing number as the single reference identification in cases where the part(s) delineated thereon are produced by the subcontractor. Prime contractor shall not add his drawing number to the drawing except as an unofficial reference outside the drawing border or margin.

3.7.1.5.1 Prime contractors who purchase semi-finished parts from subcontractors for final production phases in their own shops, have the option of using as the single drawing identification either their own title block and drawing number or the title block and drawing number of the subcontractor, but not both.

3.7.1.6 Drawings of drivers and associated equipment shall be in accordance with the specifications covering these equipments.

3.7.2 Technical manuals. In addition to the general requirements for technical manuals (see 6.2.2), the features specified in 3.7.2.1 through 3.7.2,4 shall be included.

3.7.2.1 Technical manuals shall be furnished unless otherwise specified (see 6.2.1). A separate manual shall be furnished for each different pump unit except in special cases when manuals covering more than one pump may be approved. Preliminary manuals shall be submitted to the design review agency for review and shall include all proposed sections complete. Unless otherwise specified (see 6.2.1) , manuals shall include pump, driver, controls, and appurtenances and applicable drawings. Performance curves for final manuals may be furnished as insert pages after delivery of hardware.

3.7.2.2 Manuals covering pumps driven by turbines or engines shall include drawings and full operation and maintenance instructions covering the turbines, engines governors, and any other applicable accessories. Manuals shall contain the following:

(a) Lubrication schedule inducting lubricant and procedure.
(b) Detailed seal instruction. Minimum allowable leakage rate shall be specified.

3.7.2.3 Manuals covering pumps driven by conventional electric motors with conventional controllers need not include operation and maintenance instructions covering the motors and controllers. They shall, however, include master drawings and certification data covering the motors and controllers. Complete operation and maintenance instructions covering electrical equipment shall be included when specifically required (see 6.2.1) .

3.7.2.4 Each manual shall include not less than the following illustrations covering the pump and driver:

(a) Sectional assembly drawings.
(b) Outline drawings.
(c) Complete list of material corresponding to the sectional assembly drawings.
(d) Certification data.
(e) A minimum of two photographs of the complete unit (pump and driver) taken 180 degrees apart on a horizontal plane. Where several sizes of identical design units are being supplied. photographs of one design unit only need be supplied.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, tile contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208 (see 6.2.1).

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification (see 4.3).
(b) Tests (see 4.5):
(1) Shop tests. To be conducted on each pump supplied under the contract or order (see 4.4.1).
(2) Performance evaluation tests. To be conducted on one pump of each class, design, and size on each contract or order (see 4.4.2) .
(3) Design evaluation tests. To be conducted on-one pump of each class, design, and size (see 4.4.3).

4.3 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to NAVSEC. Qualification tests shall consist of the tests specified in 4.3.1.

4.3.1 The following tests shall be performed in the order listed:

(a) Performance evaluation tests as specified in 4.4.2.
(b) Design evaluation tests as specified in 4.4.3.1 through 4.4.3.10.1.
(c) Maintenance engineering analysis shall be conducted in accordance with MIL-M-24365.

4.3.2 Qualification shall apply separately to each class. Qualification will include all sizes for a specific class.

4.3.3 Pumps submitted for qualification tests shall have characteristic ratings of not less than the values specified in table V.

TAELE V. Qualification test characteristics.

Discharge gage pressure	Rated capacity	Suction lift at rated capacity	Viscosity/fuel
(Lb/in^2)	(Gal/min)	(In Hg)	(SSU)
400-1100	60	5	35 (JP-5)
400-1100	60	10	100 (DFM)

4.3.4 Units submitted for qualification testing shall be operated at the maximum rubbing speed for which the unit is designed. Operating at a rubbing speed greater than that at which the tested unit operated on qualification test, may require an additional test subject to NAVSEC design review (see 4.4.2 . The test specified in 4.4.2 is intended to demonstrate the reliability of the pump design.

4.3.5 Unless otherwise arranged between the Government and the contractor, the pump submitted for the qualification testing shall be furnished complete with a prime mover and necessary accessories. The prime mover need not be made to Government specifications. Qualification of a pump will not include qualification of a driver.

4.3.6 The potential contractor desiring qualification of a specific pump class shall submit drawings to NAVSEC for review as a prerequisite to the performance of the qualification tests specified herein.

4.4 Tests. Failures, deficiencies, and discrepancies revealed during the performance of the specified tests and the corrective measures taken shall be recorded and fully documented in the applicable test records and test reports. After correction of deficiencies, tests shall be repeated to the extent necessary to ascertain acceptability of the modified pump. Major failures indicative of the design deficiency (as distinguished from shop error or faulty workmanship) shall be reported to the design review agency before a correction is made. Except where specifically required herein to be tested with the actual driver, tests may be conducted using a substitute driver, provided the actual driver separately passes the tests required by its applicable specification.

4.4.1 Shop tests. Shop tests shall be performed on each pump being supplied (and subcomponents of each pump if applicable). Each shop test shall be fully documented by a prescribed test procedure and a written record of the performance and findings of each test. A compilation of the documentation of the shop tests performed on each pump shall be furnished in duplicate to the cognizant Government quality assurance office and to the procuring agency. The following shop tests shall be performed on each pump:

(a) Hydrostatic pressure test (see 4.4.1.1).

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(b) Mechanical soundness and capacity tests (see 4.4.1.2).
(c) Noise test (to be performed only when specified) (see 4.4.1.3).

4,4.1.1 Hydrostatic ressure test. Each pump casing shall be tested hydrostatically to a pressure one and one-half times the maximum discharge gage pressure but in no case less than 50 lb/in^2. Hydrostatic test shall be maintained for at least 30 minutes or longer as necessary for examination of entire casing.

Acceptance criteria: Pumps shall exhibit no leakage through the casing material or joints,

4.4.1.2 Mechanical soundness and capacity tests . This shop test shall be conducted, recorded, and reported in accordance with the rotary pump Pating Standard and Test Code of the Hydraulic Institute, to the extent that these standards are applicable and are not in conflict with the requirements specified herein. The test record for each pump shall identify the pump by manufacturer's serial and model numbers, i.e., it shall indicate pump size of internals governing capacity and pressure being tested in the pump, it shall include a description (by legend. and sketch) of the test stand arrangement including an identifying list of test instruments used and the data when last calibrated, and it shall include the data sheets of all recorded data and sample calculations of data conversion to specified conditions . The test shall be performed as follows: Operate the pump, and its driver if motor driven, continuously for at least 30 minutes, to verify the ability of the pump to operate in conformance with contract requirements. Pumps shall be tested to determine their ability to deliver rated capacity using the specified fuel . This is to be performed against rated discharge pressure with the minimum specified suction lift.

Acceptance criteria: Oil temperature rise in force-feed lubricated bearings shall not exceed 50°F in any bearing and shall not exceed a maximum temperature of 180°F and the criteria specified in 4.4.3.4 (b), (c), and (d).

4.4.1.3 Noise test. When specified (see 6.2.1), a noise test shall be conducted and reported in accordance with MIL-STD-740 (see 3.3.3.5) . The noise test stand details, instrumentation, and testing technique shall be submitted by the contractor for review by the procuring agency prior to test commencement. Noise tests shall be performed with the driver furnished with the pump. Airborne noise tests shall be conducted on the lead unit only on each contract or order.

Acceptance criteria: Unit shall meet the noise level limits specified (see 6.2.1).

4.4.2 Performance evaluation tests. one pump of each class, design, and size on each contract or order, shall successfully undergo the performance evaluation tests to establish the complete pump performance map and to ascertain compliance with the specified performance requirements . A substitute driver may be used for turbine driven pumps. When pumps of identical design and size are being produced under two or more contracts within a two year period. only one series of performance evaluation tests shall be required for that production run. The proposed test procedures for the performance evaluation tests shall be submitted to the procuring agency approximately at the time of drawing submittal. The following tests shall comprise the performance evaluation tests:

(a) performance test (see 4.4.2.1).
(b) Packing/seal and alinement procedure test (see 4.4.2.2) .

4.4.2.1 Performance test. Each test pump shall be operated initially to determine pump characteristics with each of the following fuels: DFM and JP-5. Pump shall be operated with rated discharge pressure, suction temperature, and specified suction lift held constant, and flow varied from rated to minimum flow in discrete intervals (not less than 5) to develop characteristic curves for each fuel. Pump shall be operated a minimum of 1/2 hour at each test point to collect data after all parameters have stabilized. Data collected shall be used for compilation of the pump performance map and shall include efficiency and horsepower curves.

Acceptance criteria: Each test pump shall meet rated conditions with a flow capacity tolerance of plus 10 percent minus 10 percent.

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4.4.2.2 Packing/ seal and alinement procedure test. Each test pump shall have seals installed and pump an river alined in accordance with the procedures to be used in the technical manual. Seal shall then be run in and the pump operated at maximum rated speed. Test shall be conducted separately with only the mechanical seal installed and again with only the packing installed. Tests shall verify seal installation and backup packing installation procedures. Stuffing box leakage for each separate installation shall be determined.

Acceptance criteria: Pumps shall exhibit no leakage through the casing material or joints, Leakage rate through a mechanical seal shall not exceed five drops per minute.

4.4.3 Design evaluation tests. One pump of each class, design, and size submitted for qualification testing shall be provided complete with the driver and appurtenances and controls shall successfully undergo the design evaluation tests specified herein. Design changes which in the opinion of the design review agency may detrimentally affect the reliability of a previously tested and accepted pump design shall be cause to require new design evaluation tests in part or in full. In shall be at the discretion of the design review agency to require new design evaluation tests when an accumulation of several design changes, each of itself apparently not significant to pump reliability, collectively considered render suspect the validity of any one or all of the previously performed design evaluation tests. A contamination test need not be performed provided that the manufacturer can demonstrate to the satisfaction of the design review agency that the pump to be offered has been proven satisfactory in previous Naval shipboard service under conditions equally as strenuous as those specified in the ordering data. The proposed design evaluation test procedure shall be submitted to the procuring agency for review prior to performing the tests. The design evaluation tests shall consist of the following:

(a) Special operating test (see 4.4.3.1 through 4.4.3.6),
(b) Contamination test (see 4.4.3.7).
(c] Shock test (see 4.4.3.8),
¿) Inclined operation test (see 4.4.3.9).
(e) Vibration test (to be performed only when specified) (see 4.4.3.10).

4.4.3.1 Maximum fuel oil temperatures (using grade JP-5). Each test pump shall be operated at rated conditions and the oil temperature increased to 125°F at the pump discharge. Oil temperature shall be maintained at 125°F and the pump shall be operated for a period of 2 hours.

Acceptance criteria: Same as 4.4.3.4.

4.4.3.2 Suction lift test. Each test pump shall be operated at rated conditions with grade JP-5 fuel and the suction valve shall be closed until a suction pressure of 15 in IIg vacuum is attained. Pump shall maintain stable operation for 15 minutes.

Acceptance criteria: Flow shall not deteriorate by more than 50 percent and shall meet the criteria of 4.4.3.4 when valve is reopened. Pressure shall not fluctuate or discharge by more than 40 percent during test.

4.4.3.3 Minimum flow/stability test. Each test pump shall be operated with grade JP-5 fuel to determine minimum flow/stability parameters at 95°F. Parameters shall be established by operating the pump at 5 gal/rein until discharge temperature stabilizes.

Acceptance criteria: Same as 4.4.3.2 and in no case shall the discharge temperature be allowed to exceed 125°F.

4.4.3.4 Cavitation test. Each test pump shall be operated at rated conditions using grade JP-5 fuel. Suction pressure shall be reduced until cavitation is noted, then reduced an additional 2 in Hg and operated for 30 minutes in this condition. Upon completion of this test the pump shall be operated at rated conditions and the performance characteristics recorded.

Acceptance criteria:

(a) Same as 4.4.2.1.
(b) A failure is the condition where the test pump will not contain, completely, the test fluid under any operations.
(c) A failure is the condition where the test pump will not maintain the required suction lift at rated conditions.

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(d) A failure is the condition where the structureborne vibration measurement of the test pump exceeds 115 velocity decibels (VdB) as defined in MIL-STD-167-1.

Included in the above criteria is a condition where a problem is identified at an early stage and the test pump shut down, but would lead to failure if allowed to continue. Any failures that occur during the base line test shall be fully analyzed and corrected to the satisfaction of NAVSEC prior to resumption of testing. Occurrence of a failure as defined above shall result in a complete retest.

4.4.3.5 Run dry test. Each test pump shall be operated at rated conditions using grade JP-5 fuel. The suction valve and vent pump suction shall be closed to allow the pump to run dry. Continue to run the pump for 90 seconds after the flow of oil has ceased, then stop pump. Upon completion of this test, the pump shall be operated at rated conditions and the performance characteristics recorded.

Acceptance criteria: Same as 4.4.2.1.

4.4.3.6 Air ingestion test. Each test pump shall be operated at rated conditions using grade JP-5 fuel. During operation, air shall be injected into the pump suction at the rates of 25 percent by volume and the pressure fluctuation shall not exceed plus or minus 20 percent.

Acceptance criteria: Same as 4.4.3.4.

4.4.3.7 Contamination test. Test agenda and test system design details shall be submitted by the testing agency to NAVSEC for review and approval. A contamination sampling plan shall be included as part of the test agenda showing sample tap locations. and identifying sampling and analysis techniques. Approval of the system and acceptance of the testing procedure shall be prerequisite to certification of the agency for the performance of the contamination test.

4.4 ,3.7.1 Testing shall be performed using JP-5 fuel, mixed with 1.125 percent of sea water by volume. The volume of fuel in the system shall be set to provide for one complete fuel change in 3 minutes when installed pumps are at rated conditions. Test operation shall consist of each pump operating for 1000 hours at constant speed with rated conditions. System shall be contaminated with 30 mg/L using the following particulate mix:

Particulate matter (pyritic silicate) :

Initial particulate size:	25 to 50 micrometers -	10 percent by weight
	50 to 100 micrometers -	15 percent by weight
	100 to 500 micrometers -	75 percent by weight

4.4.3.7.2 Operation of the test shall be continuous throughout an initial 40 hour run-in period and during each of the forty 24-hour contaminant cycles. Each 24-hour cycle shall provide for the contaminant charging, operation, and filtering periods. Charging period shall be completed in the first hour by the slow injection of four equally spaced charges. To determine the charge weight for RACH injection, following expression shall be used:

Injection charge (grams) = 0.0075 X fuel test volume (Liters).

During the operation period the average contaminant in the flow stream at the suction side of each test pump shall not fall below 20 mg/L. In the last hour of the cycle, filtering shall be performed and provide a background level less than 5 mg/L before recharging at the beginning of the next cycle.

Acceptance criteria:

(a) A failure is the condition where the speed or torque required by the test pump to meet rated conditions has increased by 20 percent above the base line measured during the performance test specified in 4.4.2.1.
(b) Same as criteria (b) through (d) of 4.4.3.4.

4.4.3.7.3 Prior to commencement of the contamination test and. immediately after completion of the contamination test, the pump shall be disassembled to the extent necessary and the critical dimensions and. running clearances of parts subject to wear, erosion, and derangement shall be measured, calculated, and recorded. Components such as pump rotors, thrust plates, side plates, wear blocks, and casings, etc., subject to corrosion and wear, the effects of which are not subject to routine measurement, shall be listed in the inspection record, and after completion of the test the condition of each component determined and recorded,

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4.4.3.7.4 Pump shall be monitored during the endurance test to accurately record the conditions of operation, speed at which operated, and the general performance observed. Data shall be collected and the pump examined at least twice per day of operation. For each periodic examination, in addition to measured data, the record shall indicate the following:

(a) Pump packing or seal leakage rate.
(b) Conditions of the bearings.
(c) Airborne noise level (normal-abnormal).
(d) Vibration level (normal-abnormal).
(e) Smoothness of operation (normal-abnormal).
(f) Other abnormal findings.
(g) Adjustments made.
(h) Changes made in the conditions or method of operation.

4.4.3.7.5 Each test pump shall be disassembled and inspected during the test when variation of the pump parameters indicate some problem in the pump internals.

4.4.3.7.6 Post contamination test procedure. Unit subjected to the 1000 hour contamination test shall be restored to the as-new condition by replacement of parts worn beyond the as-new design tolerances. Restored unit shall successfully pass the shop tests specified in 4.4.1.1, 4.4.1.2, and 4.4.1.3, if applicable. Shop test documentation shall indicate that the unit was subjected to the contamination test and subsequently restored and shop tested, and that it shall be certified as fully conforming to the specification for unrestricted service.

4.4.3.8 Shock test. Pump shall undergo a shock test to ascertain that the pump has the necessary shock resistance. Shock test shall be performed in accordance with MIL-S-901 and the specific shock test requirements (see 6.2.1).

4.4.3.8.1 Before and after the shock test, the pump and driver and other components susceptible to internal derangement shall be disassembled to the extent necessary and the critical dimensions and running clearances measured, calculated, and recorded. During this disassembly, the critical components and assemblies subject to shock damage and derangement shall be Identified and listed in the inspection record and after completion of the test, the condition of each component and assembly determined and recorded.

4.4.3.8.2 Before and after the shock test, shop tests in accordance with 4.4.1.2 shall be performed to determine the changes in performance characteristics of the pump. Vibration measurements shall be taken at the bearing caps or housings of the pump and driver at equal speeds during the initial and final capacity test to determine the changes in mechanical operation.

4.4.3.8.3 Unit shall be mounted on the shock machine or barge essentially identical to the actual shipboard installation. The procuring agency will furnish the contractor a drawing of the shipboard mounting arrangement. Horizontal pumps when tested in the inclined position on the medium-weight shock machine shall be oriented so that the direction of shock is perpendicular to the axis of pump rotation. Pump shall be in operation during the first and third blow in each orientation on the medium-weight shock machine. Pump shall be in operation during the first, third, and fifth blows of the shock test on the floating barge. Variable speed pumps shall be operated at the minimum speed and pressure required to insure lubrication of bearings and wearing parts. Other pumps shall be operated at the lowest rated speed.

4.4.3.8.4 Pump shall be carefully observed during each shock blow and thoroughly examined visually after each blow. After each blow the unit shall be operated at as close to maximum rated speed as possible and checked for abnormal noises and vibrations and proper functioning of controls. Turbine driven pumps may be air driven.

4.4.3.8.5 Shock test acceptance criteria. Shock test acceptance criteria shall be as follows:

(a) There shall be no breakage of parts, including mounting bolts.
(b) There shall be no distortion or derangement of any part which would render the unit incapable of performing as specified.
(c) Amplitude of vibration after test at maximum rated speed shall be less than twice the amplitude measured at the same speed before the test.
(d) Adequate lubrication to bearings shall be maintained.
(e) Critical dimensions and running clearances have been maintained.
(f) There shall be no significant change in the performance curve.

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4.4.3.8.6 Post-shock-test procedure. The shock tested unit, if it is to be supplied under a contract or order and if approved by the procuring agency, shall be restored to the as-new condition by replacement of parts damaged or distorted beyond the as-new design tolerances. Rolling contact bearings shall be replaced regardless of condition. Shock tested rolling contact bearings shall be destroyed. Restored unit shall successfully pass the shop tests specified in 4.4.1.1, 4.4.1.2, and 4.4,1.3, if applicable. Shop test documentation shall indicate that the unit was subjected to the shock test and subsequently restored, shop tested, and examined. A complete part examination check list shall be supplied and shall identify the parts which were replaced (such as the bearings) and shall certify that the unit fully conforms to the specifications for unrestricted service.

4.4.3.8.7 Pump units shall be shock tested with drivers. Flexible coupled pumps shock tested with one driver will not be required to be shock tested again when supplied with a different driver of equal or less weight subject to approval of the design review agency. Prime movers shall be subject to shock tests in accordance with the applicable equipment specifications (see 6.2.1).

4.4.3.9 Inclined operation test. Vertical pumps shall be operated for not less than 30 minutes inclined at an angle from the normal equal to the combination of the maximum permanent list and trim as specified (see 6.2.1).“ Horizontal pumps shall be tested in accordance with the inclined operation test specified for the driving unit. Operation in the inclined position shall be as close to rated speed and capacity as practicable. Performance point shall be indicated in the test report. Pump shall meet all specified performance requirements without damage while operating in the inclined position.

4.4.3.10 Vibration test. When specified (see 6.2.1), one pump of a quantity of identical pumps being supplied shall successfully undergo a vibration test in accordance with the requirements of MIL-STD-167-1 and as supplemented in the contract or order. Vibration test need not be repeated on subsequent contracts or orders for pumps of identical design to those previously tested, provided the previous tests included the frequencies specified (see 6.2.1).

4.4.3.10.1 Unit shall be mounted on typical shipboard foundations during the vibration test or the shipboard mounting arrangement shall be simulated in springmass characteristics except where this mounting arrangement causes the largest test table capacity to be exceeded. Inability to vibration test the unit because of excessive weight or size shall not release the contractor from furnishing equipment which can withstand the specified vibration inputs. Vibration test acceptance criteria shall be in accordance with MIL-STD-167-1.

4.5 Test procedures and test reports. The contractor shall prepare shop test procedures (see 4.4. performance evaluation test procedures (see 4.4,2), design evaluation test procedures (see 4.4.3), and test reports in accordance with the data ordering documents included in the contract or order (see 6.2.2) and as specified in 4.5.1 and 4.5.2.

4.5.1 Test reports for the performance evaluation tests shall be compiled ‘into a single document. The document shall identify the pump by manufacturer’s serial numbers of pump and driver, the contract or order number under which the tests were performed, the ship or class of ships for which the pump is intended, if known, and the letter and data of test procedure approval.

4.5.2 Test reports for the design evaluation tests shall be prepared in accordance with good engineering practice. Test arrangement and procedure, and test events, the test instruments used, the measured data, and the effects, results, and observations shall be accurate and complete, and shall be presented in a professional manner and in a usable durable form. Report shall include a definite statement regarding conformance to the acceptable criteria specified herein and suitability of the unit for its intended application. Distribution of the test reports shall be as follows:

(a) Two to each procuring agency.
(b) Two to one of the following, as applicable:
  (1) Defense Contract Administration Semites Region (DCASR).
  (2) Defense Contract Administration Services District (DCASD).
  (3) Defense Contract Administration Services Office (DCASO).
(c) Two to the design review agency.

4.6 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation-packaging, and packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government pprocurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5) .

5.1 General. Unless otherwise specified (see 6.2.1), a first article sample shall be accomplished and technical data furnished in accordance with the requirements of MIL-P-16789.

5.2 Preservation-packaging, packing, and marking. The pump units, accessories, and repair parts shall be preserved-packaged level A or C, packed level A, B, or C, as specified (see 6.2.1), and marked in accordance with MIL-P-16789.

5.2.1 Special marking. When applicable (see 4.4.1.3 and 6.2.1), the marking requiremebts if MIL-STD-470 shall be applied to shipping containers.

5.3 Cushioning, dunnage, and wrapping materials.

5.3.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler, or dunnage is prohibited for materials destined for shipboard installation/stowage.

5.3.2 Level C preservation-packaging and packing. When loose-fill type materials are used for packaging-and packing applications such as cushioning, filler, and dunnage, containers (unit, intermediate, and shipping) shall be marked or labelled with the following information:

"CAUTION

Contents cushioned etc., with loose-fill material.
Not to be taken onboard ship.
Remove and discard loose-fill material before shipboard stowage.
If required, recushion with cellulosic material, bound fiber,
fiberboard, or transparent flexible cellular material."

5.3.3 Cushioning, filler, dunnage, and wrapping materials selected, whenever available shall exhibit improved performance for resistance to fire.

6. NOTES

6.1 Intended use. Pumps covered by this specification are intended for use on conventional fossil fuel fired steam boilers in nominal 400 lb/ın^2, 600 lb/ın^2, and 1000 lb/in^2 fuel systems.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Style and class required (see 1.2).
(c) General design requirements (see 3.3) :
  (1) Space and weight limitations (see 3.3).
  (2) List, pitch, roll, trim, and other ship attitude conditions (see 3.3 and 4.4.3.9).
  (3) Airborne, structureborne, and fluidborne noise requirements (see 3.3.3.5 and 4.4.1).
(d) Horizontal or vertical mounting (see 3.3.4.1):
  (1) If vertical, whether base mounted or table mounted (see 3.3.4.3).
  (2) Size of the inlet and outlet.
(e) Valves, flanges, fittings, bolting, and gage connections requirements (see 3.3.6.1).
(f) Suction and discharge connection (see 3.3.7.3).
(g) Whether mechanical shaft seals are required for services other than those specified (see 3.3.10.1) and whether a back-up stuffing box is required (see 3.3.10.2).
(h) Type of coupling (see 3.3.11).
(i) Type of driver (see 3.5).
(j) Turbine and motor requirements (see 3.5.1 and 3.5.4).

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(k) Type of pressure regulating governor (see 3.5.2.1).
(1) Electrical controller features, if other than as specified (see 3.5.5.1).
(m) Performance characteristics:
(1) Rated capacity and range of capacity.
(2) Rated capacity and pressure.
(3) Fluid to be pumped (see 3.3.2.2).
(4) Viscosities.
(5) Temperature of fluid, normal, maximum, and minimum (see 4.4.1.2).
(6) Suction lift.
(7) Special performance requirements, if any.
(n) Number of sets of special tools (see 3.6.2.1).
(0) Manuals required and whether complete information for electrical equipment is required (see 3.7.2.1 and 3.7.2.3).
(p) Quality assurance requirements (see 4.1.1).
(q) Noise test and noise level limits (see 4.4.1.3).
(r) Shock test requirements (see 4.4.3.8 and ordering data of MIL-S-901).
(s) Vibration test requirements (see 4.4.3.10).
(t) Ship type and fluid system in which units are to be installed.
(u) Designation of manufacturer (pump or driver) responsible to the purchaser for compatibility of pump and driver and operation of the assembled unit.
(v) First article sample and technical data requirements (see 5.1).
(w) Levels of preservation-packaging, and packing required (see 5.2) .
(x) Special marking requirements (see 5.2.1) .

6.2.2 Data requirements. When this specification is used in a procurement which invokes the provision of the "Requirements for Data" of the Armed Services Procurement Regulations (ASPR) , the data identified below, which are required to be developed by the contractor as specified on an approved Data Item Description (DD Form 1664) , and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the ASPR are not invoked in a procurement, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraphs	Data requirements	Applicable DID	Option
3.7.1 and 3.7 .1.2.4	Drawings, engineering and associated lists	DI-E-7031	Level 3 Design activity designation - Contractor Drawing No. - Contractor Delivery of hard copy - Procuring activity
3.7.2	Manual. technical, preliminary	DI-M-2043	Type I of MIL-M-15071
3.7.2	Manual, technical, standard, basic issue	DI-M-2044	Type I of MIL-M-15071
4.5	Test procedures	UDI-T-23649	
4.5	Test reports	UDI-T-23729	

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the procuring/purchasing activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item procured to this specification. This does not apply to specific data which may be required for each procurement regardless of whether an identical item has been supplied previously (for example, test reports) .

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-18526 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded

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contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of Products may be obtained from that activity. Application for Qualification tests shall be made in accordance with “Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of “Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Definitions. As used herein “procuring agency” refers to the activity purchasing the equipment covered by this specification. The procuring agency may be the Government or a shipbuilder. The “design review agency” refers to the activity responsible for controlling the technical requirements for design and testing of the equipment. The design review agency is generally a Government command or agency such as NAVSEA, NAVSEC, or an authorized representative. Communication with the design review agency should be handled through the procuring agency.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Changes from previous issue. The symbol “#” is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4320-N173)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A GIVE PARAGRAPH NUMBER AND WORDING

   B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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